CE205 Databases and Information Retrieval

Evaluation of Ranked Retrieval IR System

Class 9

Objective

The purpose of this exercise is to evaluate an IR system using various performance metrics when the retrieved documents are ranked.

Problem 1:

For the given ranking of an IR system for **query 1**, determine

- precision and recall at every rank position
- precision and recall at rank 1, 3, 6, 10 and 20.
- average precision (AP)

Now, For the given ranking of an IR system for **query 2**, determine

- precision and recall at every rank position
- precision and recall at rank 1, 3 and 15
- average precision (AP)

Also calculate Mean Average Precision (MAP).

Query 1

Rank	Relevant Document
1	X
2	
3	X
4	
5	
6	X
7	
8	
9	
10	X
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	X

Query 2

Rank	Relevant Document
1	X
2	
3	X
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	X

Problem 2

Calculate Kappa Statistic using $\text{kappa} = \frac{p(A) - p(E)}{1 - p(E)}$ for

		Judge 2 Relevance		
		Yes	No	Total
Judge 1 Relevance	Yes	300	20	320
	No	10	70	80
	Total	310	90	400